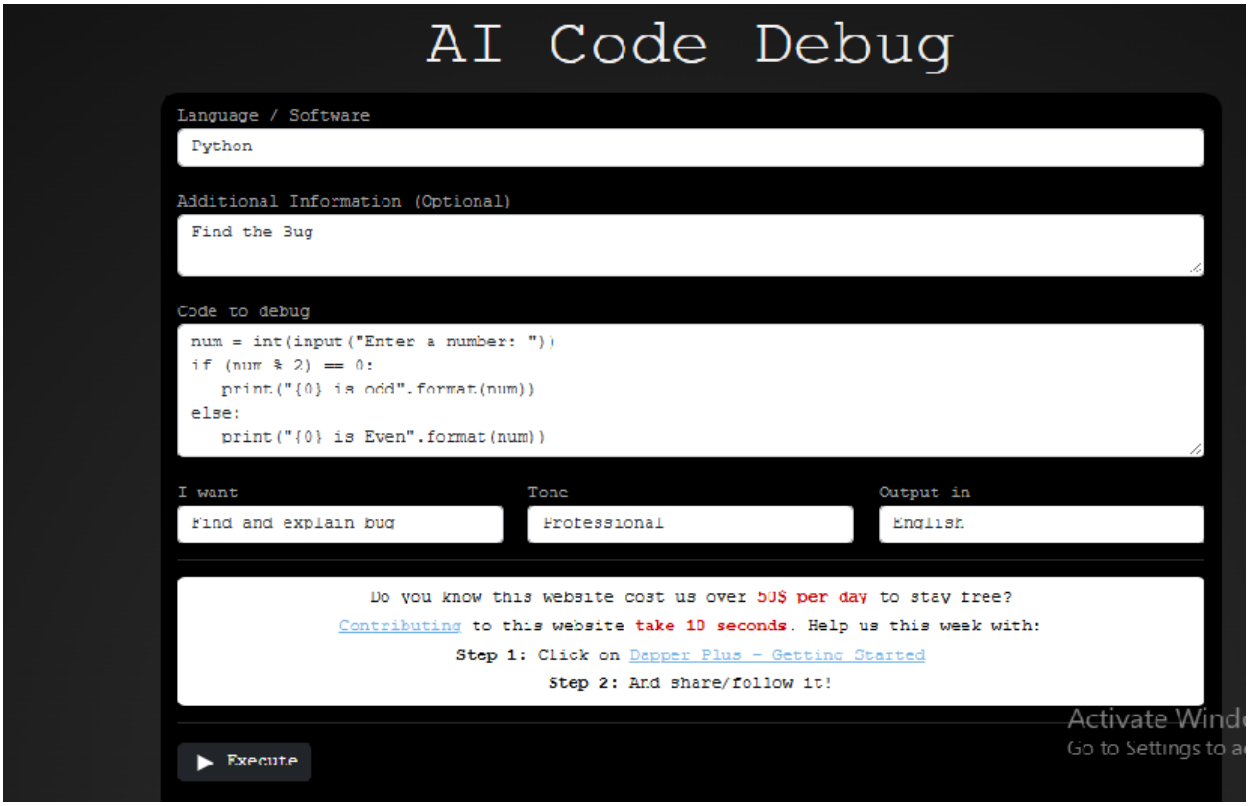
Experiment-11: Analyze the given program using Static Code Analyzer and explain its logic, structure, execution flowand output.

Aim

To analyze and understand the given program using Static Code Analyzer, and to study its logic, structure, execution flow, and output.

Procedure

1. Open the **ZZZCode.ai** website.
2. Select the appropriate programming language.
3. Enter or upload the program to be analyzed.
4. Press Execute for analysis.
5. Examine the explanation provided by ZZZCode.ai, including the program logic and execution flow.
6. Check the expected output of the program.
7. Compare the analyzed result with the actual execution of the program.
8. Note any errors or improvements suggested by the tool.



Output:

Debugging Python Code: Identifying and Fixing Errors

Code Summary

The provided Python code is intended to determine whether a given number is odd or even. However, there is a logical error in the print statements that needs to be addressed.

Problem

The code incorrectly identifies odd and even numbers due to a mix-up in the print statements. Specifically, the output messages for odd and even numbers are swapped.

Cause

In the code, the condition checks if the number is even using the modulus operator (%). If the number is even, it should print that the number is even, but the code currently states that it is odd. Conversely, if the number is odd, it incorrectly states that it is even.

Solution

To resolve this issue, we need to correct the print statements to accurately reflect whether the number is odd or even. Here is the corrected code:

language-python

Copy code

```
num = int(input("Enter a number: "))
if (num % 2) == 0:
print("{0} is Even".format(num))  # Corrected to indicate the number is even
else:
print("{0} is odd".format(num))     # Corrected to indicate the number is odd
```

Prevention

To prevent similar issues in the future, it is advisable to:

1. Review Logic: Always double-check the logic of conditions and corresponding outputs.
2. Test Cases: Implement test cases for various inputs to ensure the code behaves as expected.
3. Code Reviews: Engage in peer code reviews to catch logical errors that may be overlooked by the original author.

By following these practices, you can enhance the reliability and accuracy of your code.

Result

The given program was successfully analyzed using **ZZZCode.ai**. The program's logic, structure, execution flow, and expected output were understood, and the correctness of the program was verified.